EE1060: Differential Equations and Transform Techniques - Quiz 2

Name: ______________________________

Roll No: ______________________________

Instructions

1. The quiz will last for 45 minutes, beginning at 14:45. Please ensure that you submit your paper by 15:30. **Late submissions may incur penalties**, so make sure to return the paper on time.

2. Go through all the questions carefully. Any **misinterpretations** will **not** be considered.

3. Write all your answers in the answer book. It is essential that your responses are **neat** and **legible**. If your answers are **not clear or readable**, they will **not be graded**. You cannot later claim that there are answers written in some obscure corner of the paper that were not evaluated. Ensure your answers are clearly presented and easily identifiable.

4. After answering all the questions you wish to, **mark the end of your answer book clearly with a distinct mark, such as a line**.

---

1. (2 points) Solve the first-order ordinary differential equation

$$(x + xy)\, dx + \left(x^2\right) dy = 0 \tag{1}$$

Solution:

$$(x + xy)\, dx + x^2\, dy = 0$$

Factor and divide by x^2:

$$\frac{1+y}{x}\, dx + dy = 0$$

Separate variables:

$$\frac{dy}{1+y} = -\frac{dx}{x}$$

Integrate both sides:

$$\int \frac{1}{1+y}\, dy = -\int \frac{1}{x}\, dx \implies \ln|1+y| = -\ln|x| + C$$

2. (2 points) Solve the first-order ordinary differential equation

$$\frac{dy}{dx} - 5y = e^{-2x}y^2 \tag{2}$$

Solution:

$$\frac{dy}{dx} - 5y = e^{-2x}y^2$$

Rewriting as Bernoulli's equation:

$$\frac{dy}{dx} = 5y + e^{-2x}y^2$$

Substitute $v = \frac{1}{y}$, so $\frac{dv}{dx} = -\frac{1}{y^2}\frac{dy}{dx}$:

$$\frac{dv}{dx} + 5v = -e^{-2x}$$

Solve the linear equation:

$$\frac{dv}{dx} + 5v = -e^{-2x}$$

The integrating factor is $\mu(x) = e^{5x}$, multiply through:

$$\frac{d}{dx}\left(e^{5x}v\right) = -e^{3x}$$

Integrating both sides:

$$e^{5x}v = -\frac{1}{3}e^{3x} + C$$

Simplify:

$$v = -\frac{1}{3}e^{-2x} + Ce^{-5x}$$

Substitute $v = \frac{1}{y}$:

$$\frac{1}{y} = -\frac{1}{3}e^{-2x} + Ce^{-5x}$$

Solving for y:

$$y = \frac{3}{-e^{-2x} + 3Ce^{-5x}}.$$

3. (3 points) Consider the ODE

$$\frac{dy}{dt} = y^3 - 6y^2 + 9y \tag{3}$$

(a) (1 point) Compute the equilibrium points of the system.

Solution:

The equilibrium points of the system are $y^* = 0$, $y^* = 3$.

(b) (1 point) Derive the ODE corresponding to linearized system around the any one equilibrium point computed in 3(a) .

Solution:

At $y^* = 0$, the linearized system is $\frac{d\Delta y}{dt} = f'(y^*)\Delta y = 9\Delta y$.

At $y^* = 3$, the linearized system is $\frac{d\Delta y}{dt} = f'(y^*)\Delta y = 0$

(c) (1 point) Derive the condition on step length for with Runge-Kutta 2nd order method is numerically stable.

Solution:

The condition for step length is $\left|1 + h\lambda + \frac{h^2\lambda^2}{2}\right| < 1$. For $y^* = 0$, the condition for step length is $\left|1 + 9h + \frac{81h^2}{2}\right| < 1$. For $y^* = 3$, the method is stable for all h.

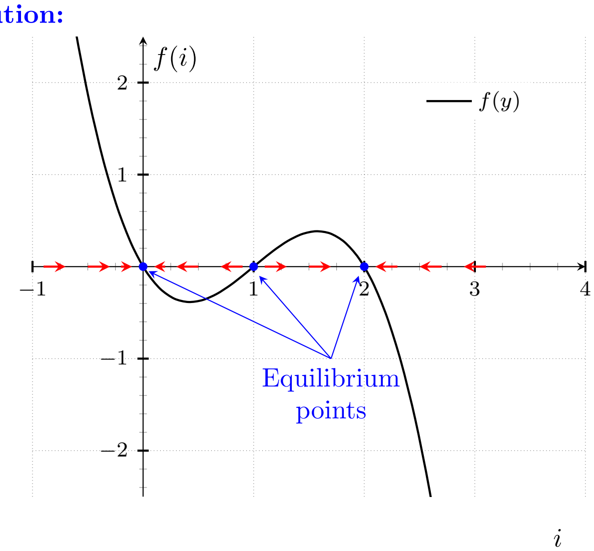
4. (3 points) Consider the ODE

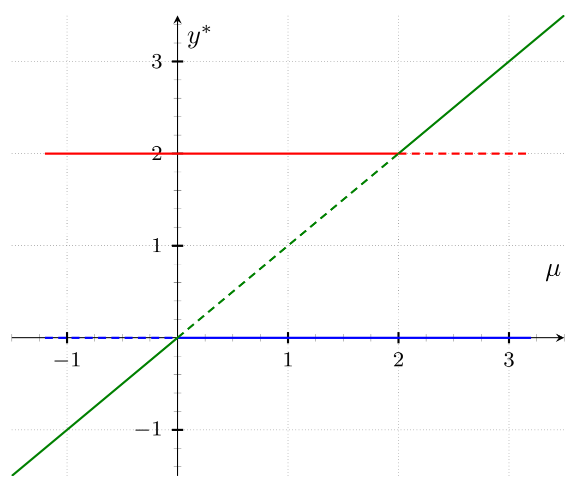
$$\frac{dy}{dt} = y\,(2-y)\,(y-\mu) \tag{4}$$

(a) (1 point) For $\mu = 1$, draw the phase plane trajectories.

(b) (2 points) Plot the bifurcation diagram for the system.

Solution:





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1. (2 points) Solve the first-order ordinary differential equation

$$(x + xy)\, dx + \left(x^3\right) dy = 0 \tag{1}$$

Solution:

$$(x + xy)\, dx + x^3\, dy = 0$$

Factor and divide by x^3:

$$\frac{1+y}{x^2}\, dx + dy = 0$$

Separate variables:

$$\frac{dy}{1+y} = -\frac{1+y}{x^2}\, dx$$

Integrate both sides:

$$\int \frac{dy}{1+y} = \int -\frac{dx}{x^2} \implies \ln|1+y| = -\frac{1}{x} + C$$

2. (2 points) Solve the first-order ordinary differential equation

$$\frac{dy}{dx} - 6y = e^{-2x} y^2 \tag{2}$$

Solution:

$$\frac{dy}{dx} - 6y = e^{-2x}y^2$$

Rewriting as Bernoulli's equation:

$$\frac{dy}{dx} = 6y + e^{-2x}y^2$$

Substitute $v = \frac{1}{y}$, so $\frac{dv}{dx} = -\frac{1}{y^2}\frac{dy}{dx}$:

$$\frac{dv}{dx} + 6v = -e^{-2x}$$

Solve the linear equation:

$$\frac{dv}{dx} + 6v = -e^{-2x}$$

The integrating factor is $\mu(x) = e^{6x}$, multiply through:

$$\frac{d}{dx}\left(e^{6x}v\right) = -e^{4x}$$

Integrating both sides:

$$e^{6x}v = -\frac{1}{5}e^{4x} + C$$

Simplify:

$$v = -\frac{1}{5}e^{-2x} + Ce^{-6x}$$

Substitute $v = \frac{1}{y}$:

$$\frac{1}{y} = -\frac{1}{5}e^{-2x} + Ce^{-6x}$$

Solving for y:

$$y = \frac{5}{-e^{-2x} + 5Ce^{-6x}}.$$

3. (3 points) Consider the ODE

$$\frac{dy}{dt} = y^3 - 6y^2 + 8y \tag{3}$$

(a) (1 point) Compute the equilibrium points of the system.

Solution:

The equilibrium points of the system are $y^* = 0$, $y^* = 2$ and $y^* = 4$.

(b) (1 point) Derive the ODE corresponding to linearized system around the any one equilibrium point computed in 3(a) .

Solution:

At $y^* = 0$, the linearized system is $\dfrac{d\Delta y}{dt} = f'(y^*)\Delta y = 8\Delta y$.

At $y^* = 2$, the linearized system is $\dfrac{d\Delta y}{dt} = f'(y^*)\Delta y = -4\Delta y$

At $y^* = 4$, the linearized system is $\dfrac{d\Delta y}{dt} = f'(y^*)\Delta y = 8\Delta y$

(c) (1 point) Derive the condition on step length for with Runge-Kutta 2nd order method is numerically stable.

Solution:

The condition for step length is $\left|1 + h\lambda + \frac{h^2\lambda^2}{2}\right| < 1$. For $y^* = 0$, $y^* = 4$ the condition for step length is $\left|1 + 8h + 32h^2\right| < 1$. For $y^* = 2$ the condition for step length is $\left|1 - 4h + 8h^2\right| < 1$.

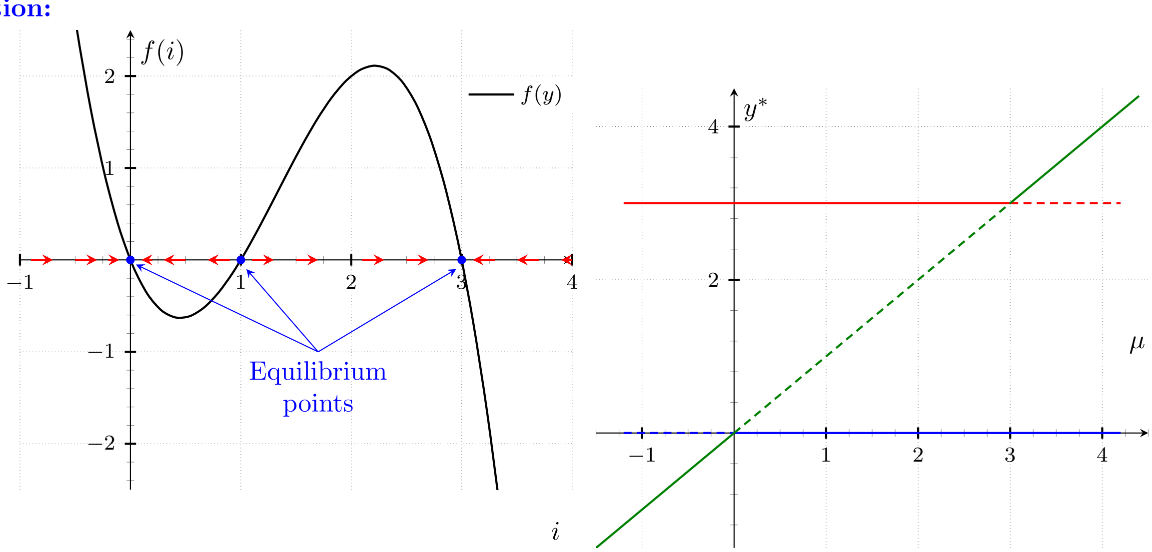
4. (3 points) Consider the ODE

$$\frac{dy}{dt} = y\,(3-y)\,(y-\mu) \tag{4}$$

(a) (1 point) For $\mu = 1$, draw the phase plane trajectories.

(b) (2 points) Plot the bifurcation diagram for the system.

Solution:



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1. (2 points) Solve the first-order ordinary differential equation

$$(x + xy)\,dx + \left(x^4\right) dy = 0 \tag{1}$$

Solution:

$$(x + xy)\,dx + x^4\,dy = 0$$

Factor and divide by x^4:

$$\frac{1+y}{x^3}\,dx + dy = 0$$

Separate variables:

$$\frac{dy}{1+y} = -\frac{1+y}{x^3}\,dx$$

Integrate both sides:

$$\int \frac{dy}{1+y} = \int -\frac{dx}{x^3} \implies \ln|1+y| = -\frac{1}{2x^2} + C$$

2. (2 points) Solve the first-order ordinary differential equation

$$\frac{dy}{dx} - 4y = e^{-2x}y^2 \tag{2}$$

Solution:

$$\frac{dy}{dx} - 4y = e^{-2x}y^2$$

Rewriting as Bernoulli's equation:

$$\frac{dy}{dx} = 4y + e^{-2x}y^2$$

Substitute $v = \frac{1}{y}$, so $\frac{dv}{dx} = -\frac{1}{y^2}\frac{dy}{dx}$:

$$\frac{dv}{dx} + 4v = -e^{-2x}$$

Solve the linear equation:

$$\frac{dv}{dx} + 4v = -e^{-2x}$$

The integrating factor is $\mu(x) = e^{4x}$, multiply through:

$$\frac{d}{dx}\left(e^{4x}v\right) = -e^{2x}$$

Integrating both sides:

$$e^{4x}v = -\frac{1}{3}e^{2x} + C$$

Simplify:

$$v = -\frac{1}{3}e^{-2x} + Ce^{-4x}$$

Substitute $v = \frac{1}{y}$:

$$\frac{1}{y} = -\frac{1}{3}e^{-2x} + Ce^{-4x}$$

Solving for y:

$$y = \frac{3}{-e^{-2x} + 3Ce^{-4x}}.$$

3. (3 points) Consider the ODE

$$\frac{dy}{dt} = y^3 - 3y^2 + 2y \tag{3}$$

(a) (1 point) Compute the equilibrium points of the system.

Solution:

The equilibrium points of the system are $y^* = 0$, $y^* = 1$ and $y^* = 2$.

(b) (1 point) Derive the ODE corresponding to linearized system around the any one equilibrium point computed in 3(a) .

Solution:

At $y^* = 0$, the linearized system is $\dfrac{d\Delta y}{dt} = f'(y^*)\Delta y = 2\Delta y$.
At $y^* = 1$, the linearized system is $\dfrac{d\Delta y}{dt} = f'(y^*)\Delta y = -1\Delta y$
At $y^* = 2$, the linearized system is $\dfrac{d\Delta y}{dt} = f'(y^*)\Delta y = 2\Delta y$

(c) (1 point) Derive the condition on step length for with Runge-Kutta 2nd order method is numerically stable.

Solution:

The condition for step length is $\left|1 + h\lambda + \frac{h^2\lambda^2}{2}\right| < 1$. For $y^* = 0$, $y^* = 2$ the condition for step length is $\left|1 + 2h + 2h^2\right| < 1$. For $y^* = 1$ the condition for step length is $\left|1 - h + \frac{h^2}{2}\right| < 1$.

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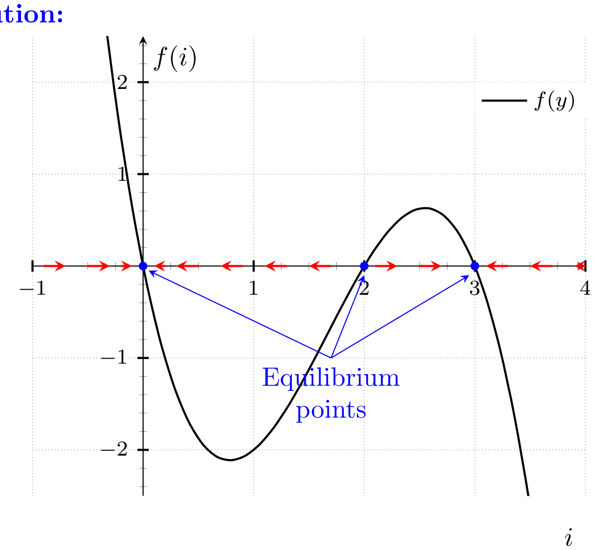

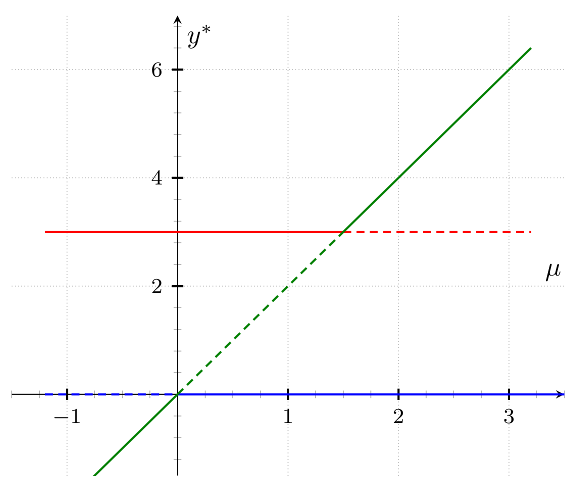

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1. (2 points) Solve the first-order ordinary differential equation

$$(x + xy)\,dx + x dy = 0 \tag{1}$$

Solution:

$$(x + xy)\,dx + x\,dy = 0$$

Factor and divide by x:

$$(1 + y)\,dx + dy = 0$$

Separate variables:

$$\frac{dy}{1+y} = -dx$$

Integrate both sides:

$$\int \frac{dy}{1+y} = \int dx \implies \ln|1+y| = -x + C$$

2. (2 points) Solve the first-order ordinary differential equation

$$\frac{dy}{dx} - 3y = e^{-2x}y^2 \tag{2}$$

Solution:

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$$\frac{d}{dx}\left(e^{3x}v\right) = -e^{x}$$

Integrating both sides:

$$e^{3x}v = -e^{x} + C$$

Simplify:

$$v = -e^{-2x} + Ce^{-3x}$$

Substitute $v = \frac{1}{y}$:

$$\frac{1}{y} = -e^{-2x} + Ce^{-3x}$$

Solving for y:

$$y = \frac{1}{-e^{-2x} + Ce^{-3x}}.$$

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